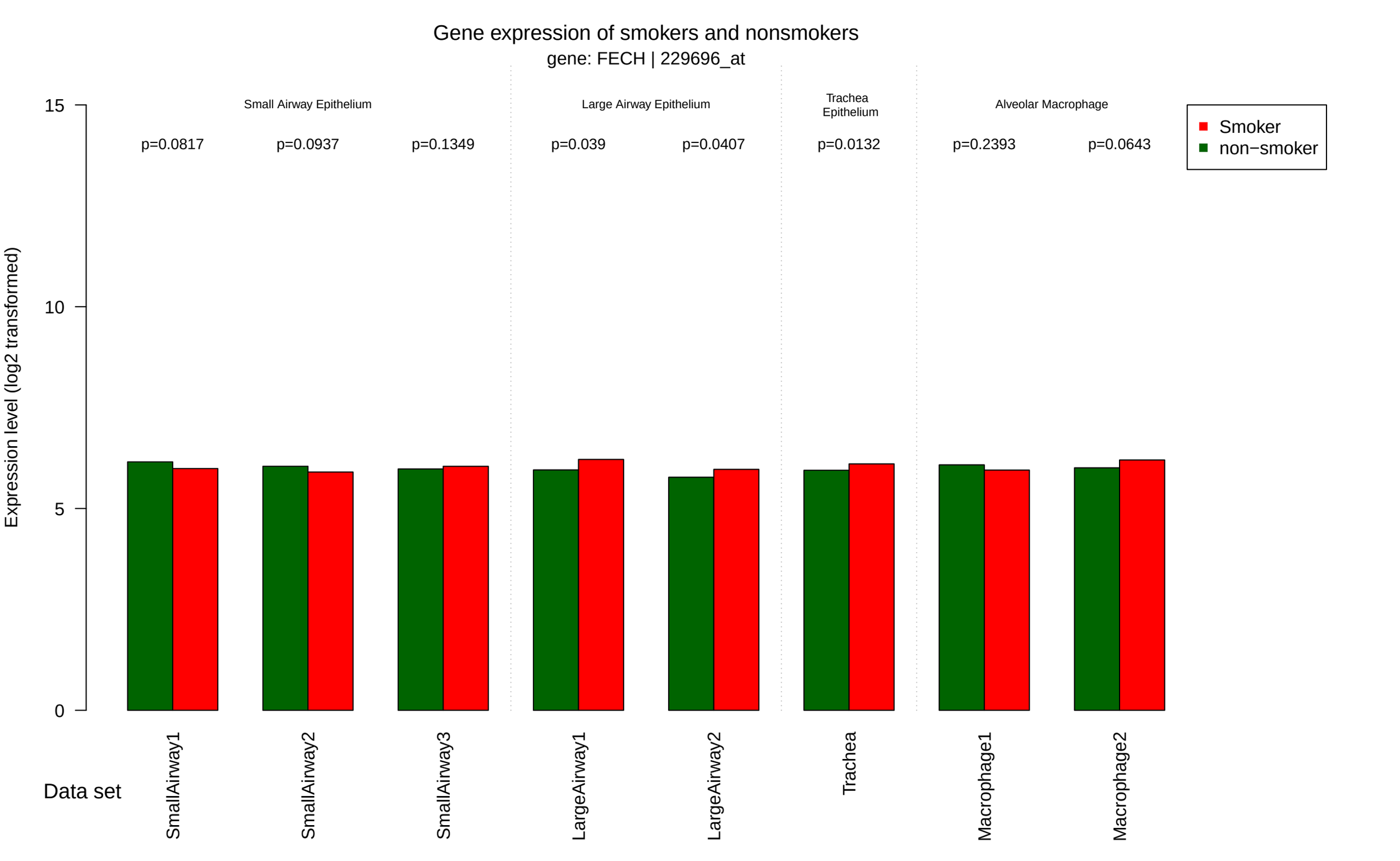
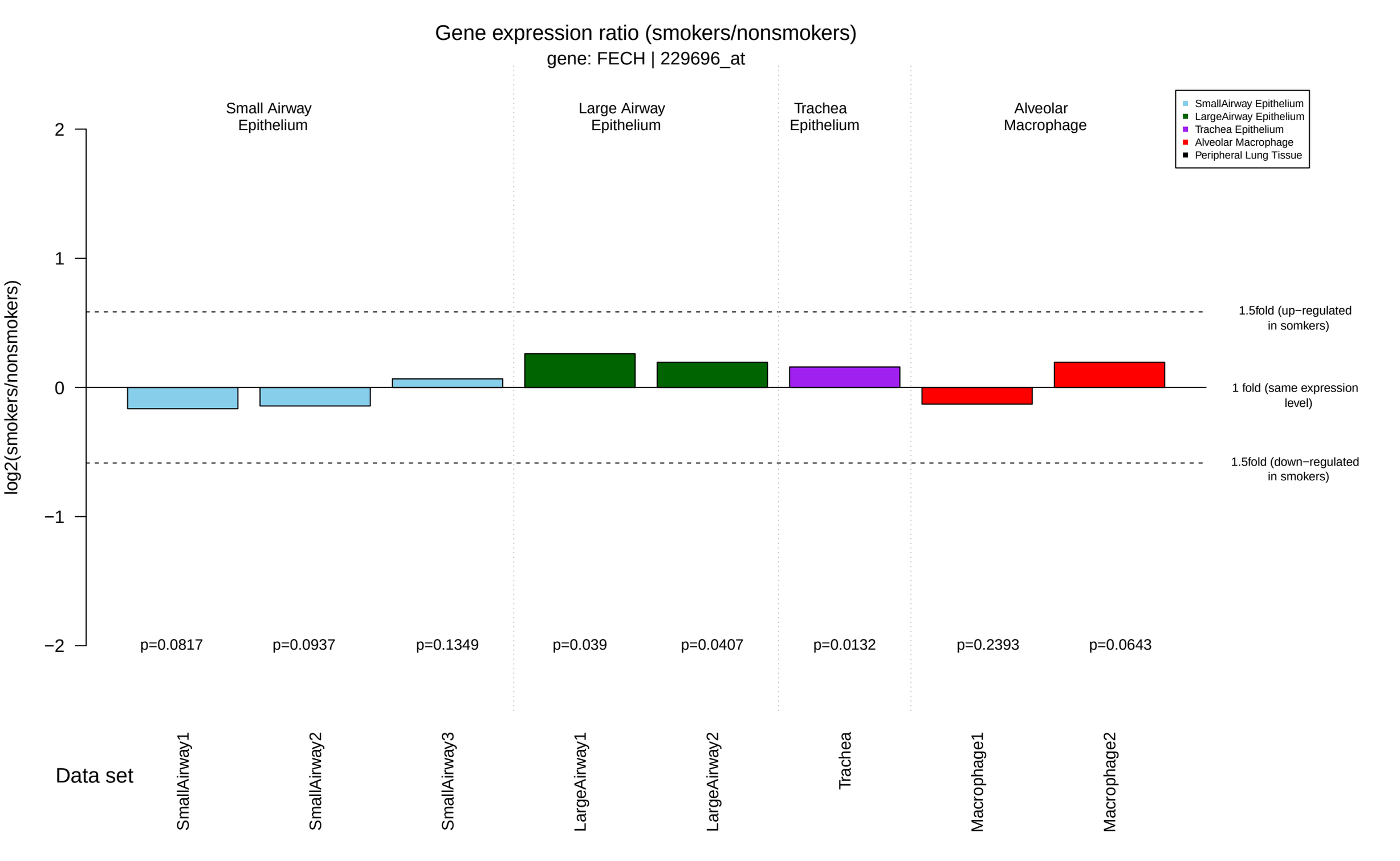
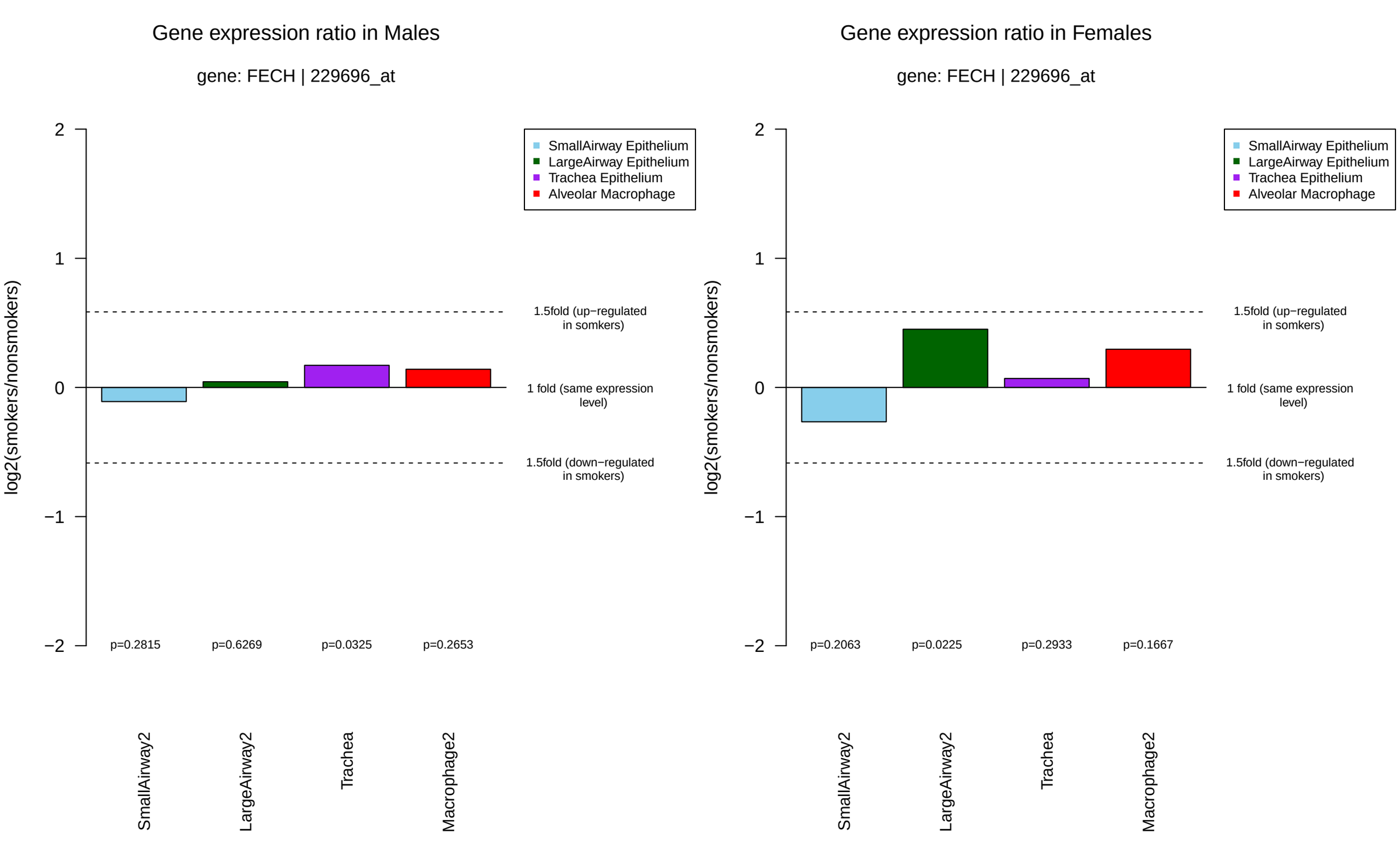
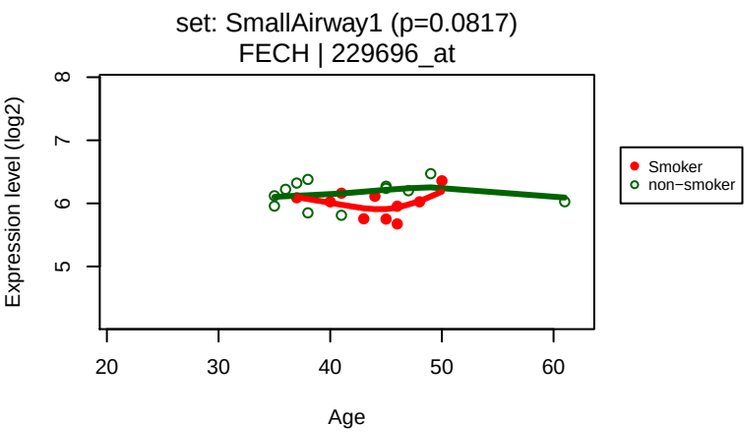
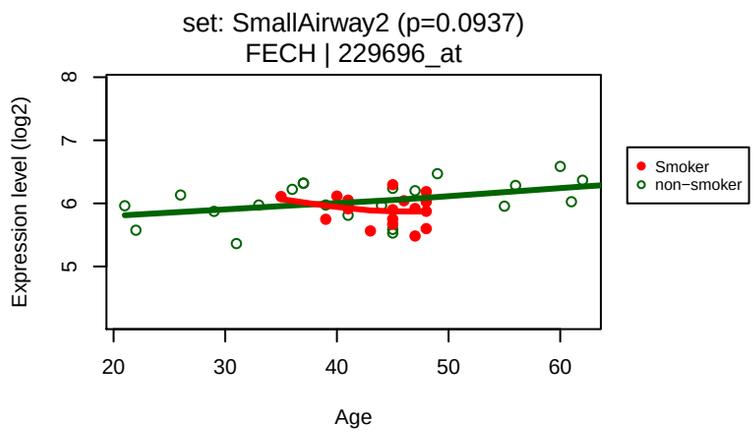
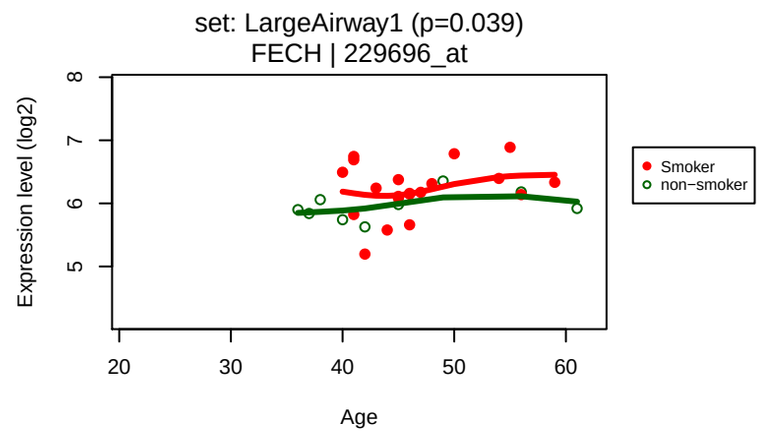

Gene expression of smokers and nonsmokers
gene: FECH | 229696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0817
p=0.0937
p=0.1349
p=0.039
p=0.0407
p=0.0132
p=0.2393
p=0.0643
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

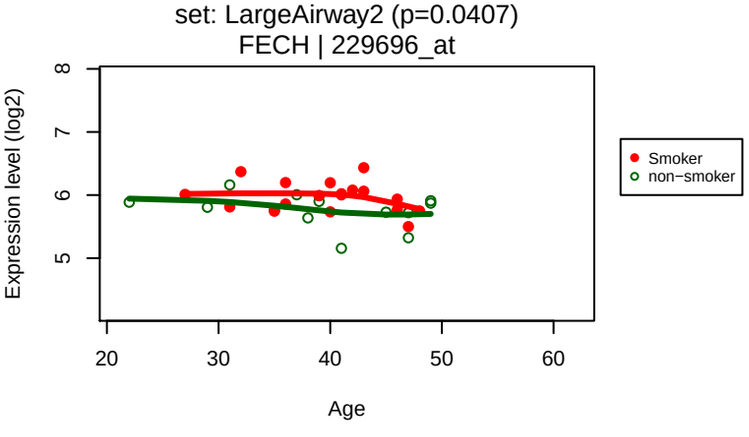

Gene expression ratio (smokers/nonsmokers)
gene: FECH | 229696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0817
p=0.0937
p=0.1349
p=0.039
p=0.0407
p=0.0132
p=0.2393
p=0.0643
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

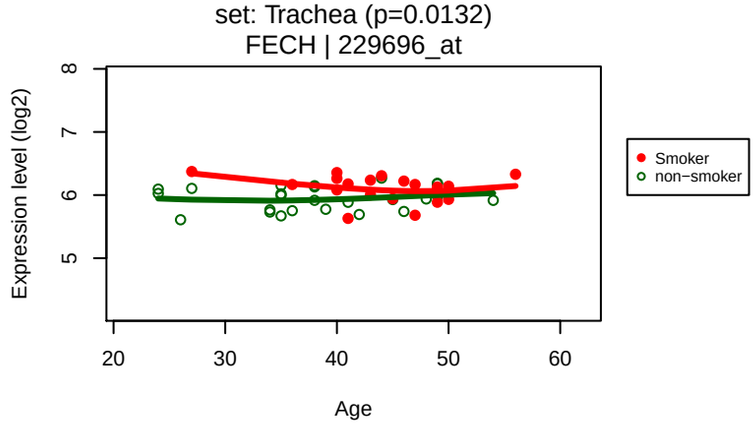

Gene expression ratio in Males
gene: FECH | 229696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2815
p=0.6269
p=0.0325
p=0.2653
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FECH | 229696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2063
p=0.0225
p=0.2933
p=0.1667
SmallAirway2
LargeAirway2
Trachea
Macrophage2

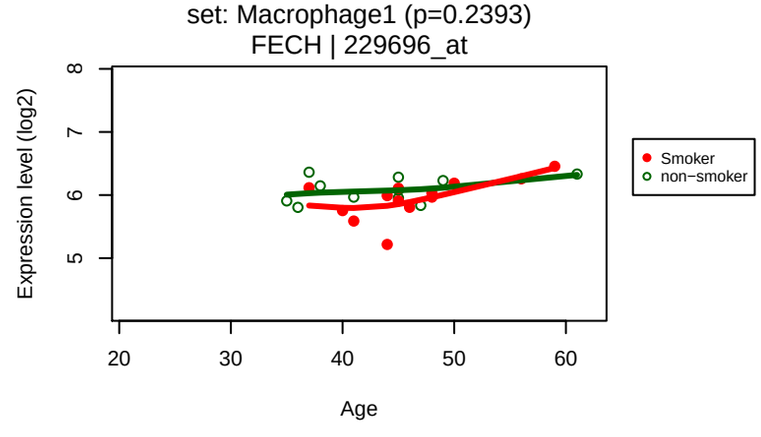

set: SmallAirway1 (p=0.0817)
FECH | 229696_at
Expression level (log2)
Age
Smoker
non−smoker

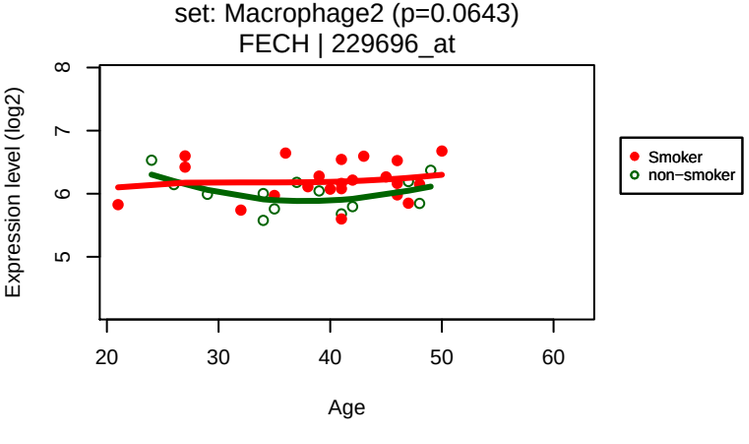

set: SmallAirway2 (p=0.0937)
FECH | 229696_at
Expression level (log2)
Age
Smoker
non−smoker

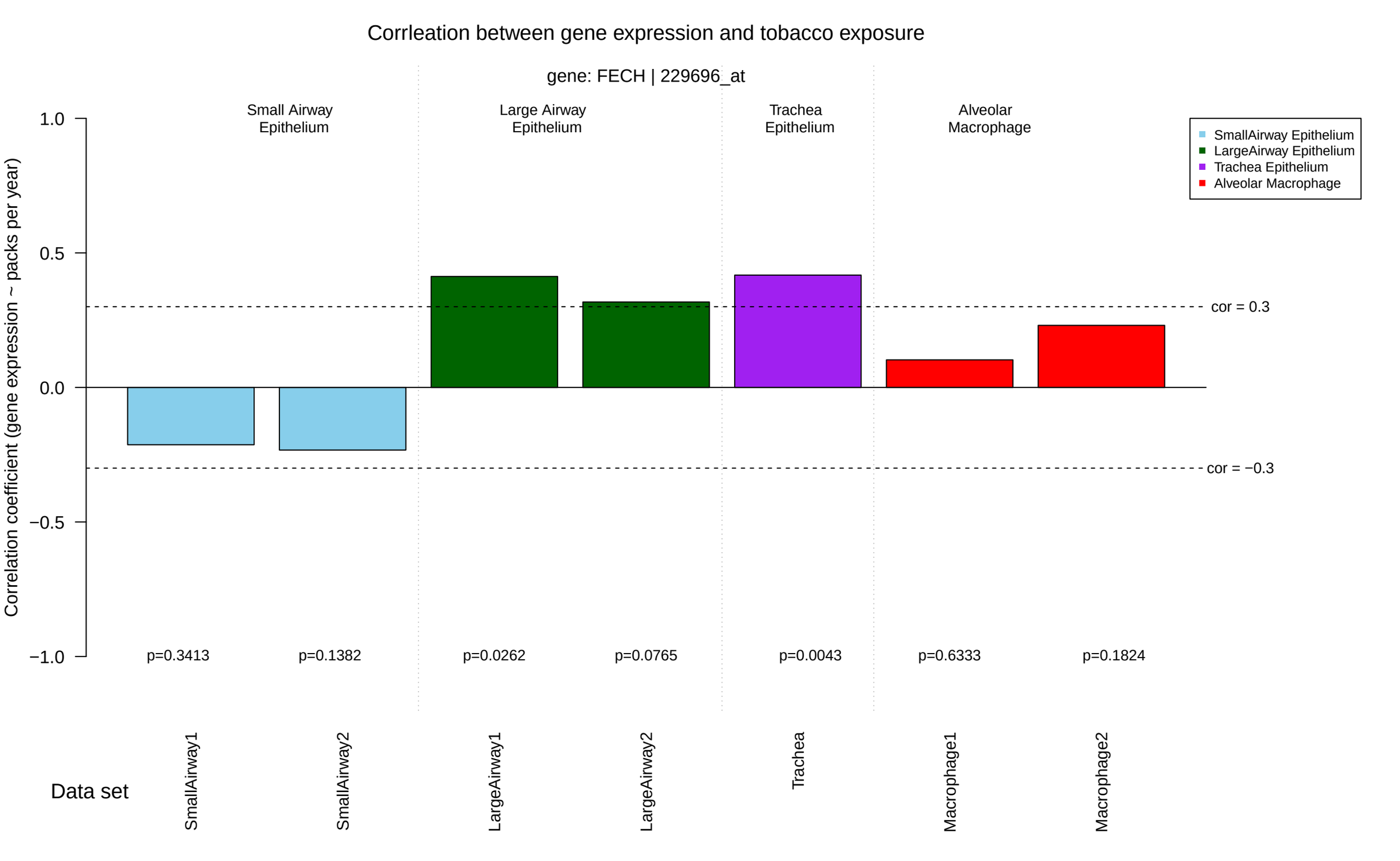

set: LargeAirway1 (p=0.039)
FECH | 229696_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0407)
FECH | 229696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0132)
FECH | 229696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2393)
FECH | 229696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0643)
FECH | 229696_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FECH | 229696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3413
p=0.1382
p=0.0262
p=0.0765
p=0.0043
p=0.6333
p=0.1824
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2